Python for Artificial Intelligence

Matplotlib & Seaborn: Visualizing Data

Week 3, Day 4

Today's Agenda

- Hour 1: Foundations (data types, distributions) and univariate analysis: bar charts and histograms
- Hour 2: Bivariate analysis: scatter plots, box plots, and reading relationships with regplot
- Hour 3: Multivariate analysis with correlation heatmaps, plus styling Seaborn plots
- By the end: you can turn a DataFrame into a chart that answers a real question

Hour 1

Foundations & Univariate Analysis

CONCEPT

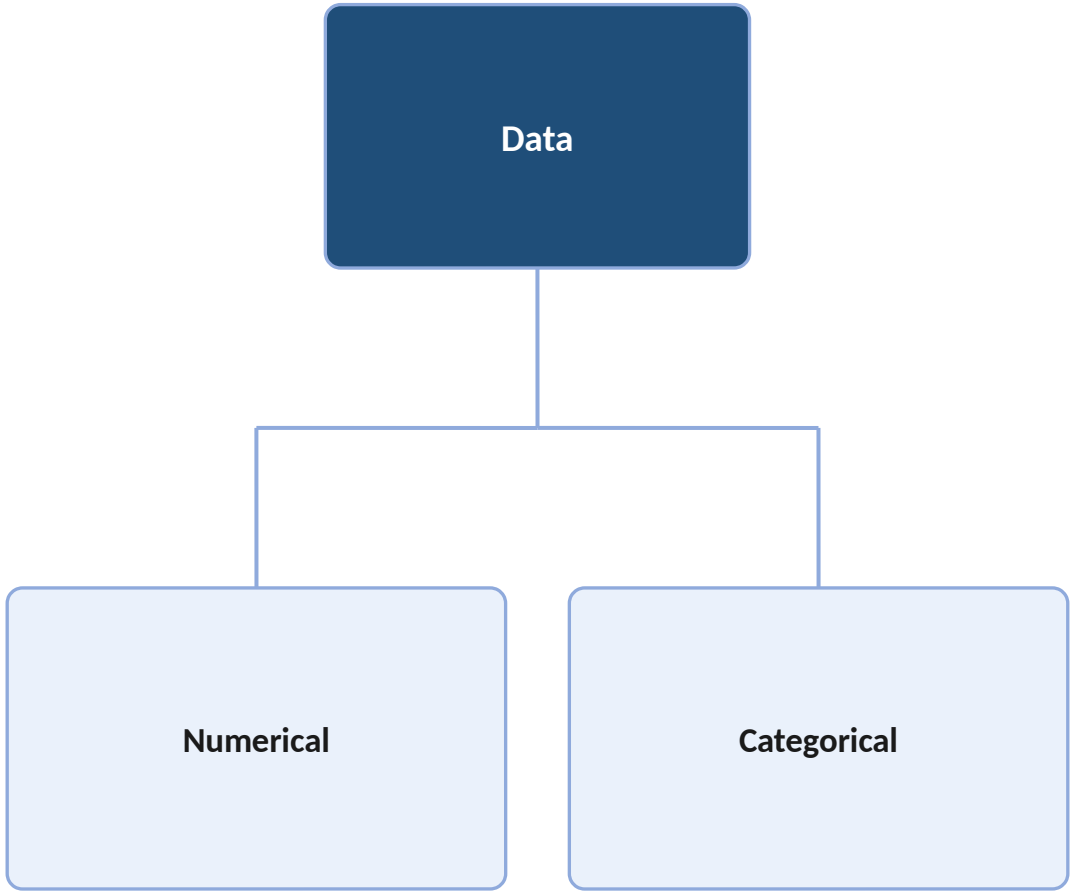
Why Visualize Data?

- A table of a thousand numbers is hard to reason about; a chart of the same numbers often isn't
- Matplotlib is Python's foundational plotting library; almost every other plotting tool builds on it
- A chart can reveal a trend, an outlier, or a relationship that .describe() alone would never show you
- The core workflow is always the same: build the plot, label it, then plt.show() it

CONCEPT

Types of Data

- Numerical (quantitative) data is measured in numbers: age, price, score, temperature
- Discrete numerical data is countable in whole steps, like number of orders or number of students
- Continuous numerical data can take any value in a range, like height, revenue, or temperature
- Categorical (qualitative) data is labels, not numbers, like city, product type, or pass/fail



CONCEPT

Nominal vs. Ordinal Categorical Data

- Nominal data has categories with no natural order: city, color, product type
- Ordinal data has categories with a real order: rating (low, medium, high), education level
- You can meaningfully say one ordinal category is "more" than another; nominal categories can't be ranked
- This distinction matters later: ordinal data can be sorted and compared, nominal data can only be counted

EXAMPLE

Data Types, Side by Side

The same dataset usually mixes all four types across its columns.

Data Type	Example Column	Example Values	Can You Average It?
Discrete (Numerical)	number_of_orders	0, 1, 2, 3 …	Yes, a whole-number average
Continuous (Numerical)	temperature	21.5, 36.9, 98.6 …	Yes, any decimal value
Nominal (Categorical)	city	Karachi, Lahore, Islamabad	No, only count or find the mode
Ordinal (Categorical)	satisfaction_rating	Low, Medium, High	No, but it can be ranked and sorted

CONCEPT

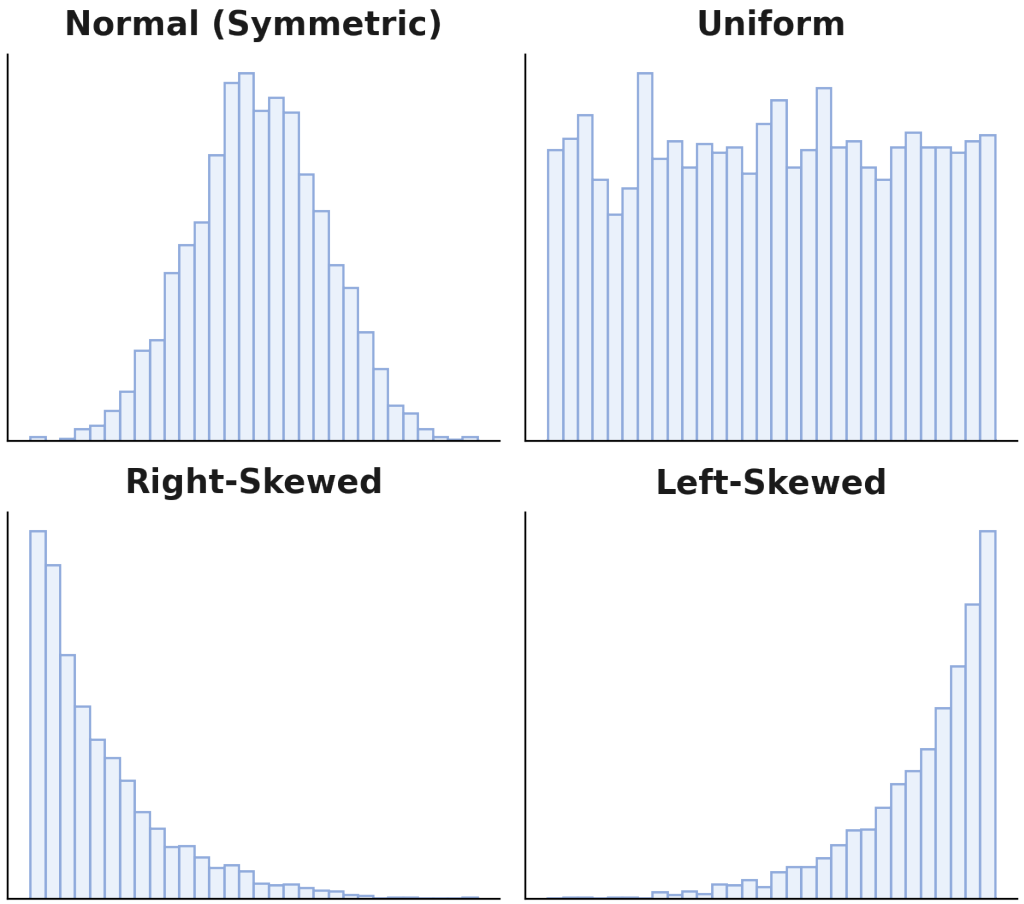
Why Check the Shape of a Distribution?

- The shape tells you whether the mean is even a fair summary, or whether the median represents it better
- Many statistical and machine learning methods assume roughly normal data; checking the shape tells you if that holds
- A skewed or multi-peaked shape can reveal subgroups or data issues that a single average would hide completely
- Recognizing the shape early saves you from over-trusting one number later in the analysis

CONCEPT

Types of Distribution

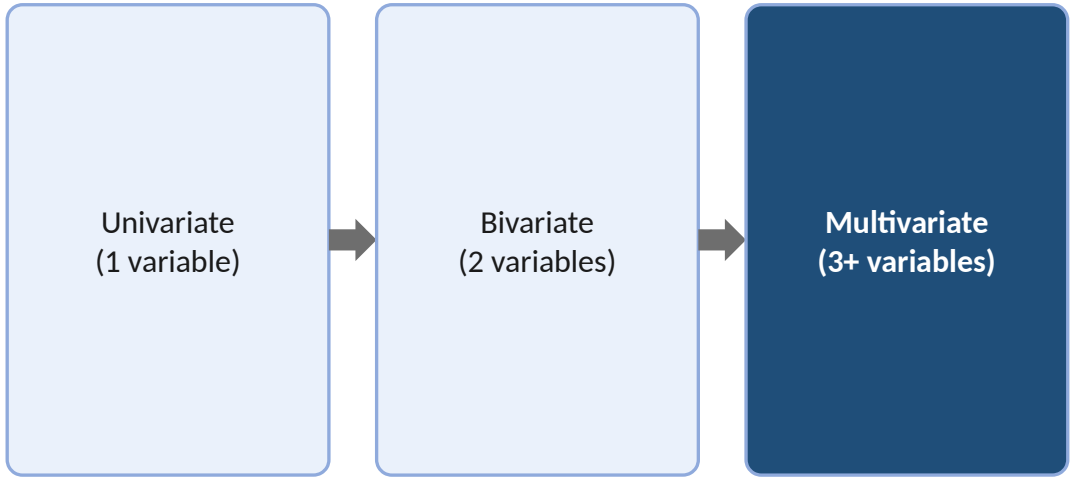
- Normal: symmetric, bunched around the mean, the classic bell shape; mean, median, and mode line up
- Uniform: every value is about equally likely, so the bars are roughly the same height across the range
- Right-skewed: a long tail of high values, like income or revenue; mean sits to the right of median and mode
- Left-skewed: a long tail of low values; mean sits to the left of median and mode



CONCEPT

Univariate, Bivariate & Multivariate Analysis

- Univariate analysis looks at one variable alone: its distribution, center, and spread
- Bivariate analysis looks at two variables together, to see whether they move with each other
- Multivariate analysis looks at three or more variables at once, often across a category
- This lecture builds up in that order: one variable, then two, then several together



Basic Plot Syntax

SYNTAX

```
import matplotlib.pyplot as plt

plt.plot(x, y)
plt.xlabel("...")
plt.ylabel("...")
plt.title("...")
plt.show()
```

PARAMETERS

`x, y` two equal-length lists or arrays: positions along each axis

`plt.show()` renders the figure; without it, nothing appears on screen

YOU CAN

✓ Swap plt.plot() for plt.bar(), plt.hist(), or plt.scatter() and keep everything else the same

✓ Call xlabel, ylabel, and title on any plot to make it self-explanatory

✓ Build several plots in the same script, one plt.show() per figure

YOU CANNOT

✗ Read a chart's meaning from code alone; you actually have to look at plt.show()'s output

✗ Plot two lists of different lengths without an error

Returns: not applicable; these calls draw onto a figure rather than returning a value you use

EXAMPLE

A Simple Line Plot

plt.plot() connects the points in order, which is why line plots suit data over time.

```
import matplotlib.pyplot as plt

days = [1, 2, 3, 4, 5]
sales = [100, 150, 130, 170, 200]

plt.plot(days, sales)
plt.xlabel("Day")
plt.ylabel("Sales")
plt.title("Sales Over 5 Days")
plt.show()
# an upward, slightly bumpy line
```

THINK & PRACTICE

Quick Check

Plot a line chart of temperature over a week: days = [1,2,3,4,5,6,7], temps = [30, 32, 31, 33, 35, 34, 36]. Label both axes and give it a title.

ANSWER

Quick Check: Answer

```
import matplotlib.pyplot as plt

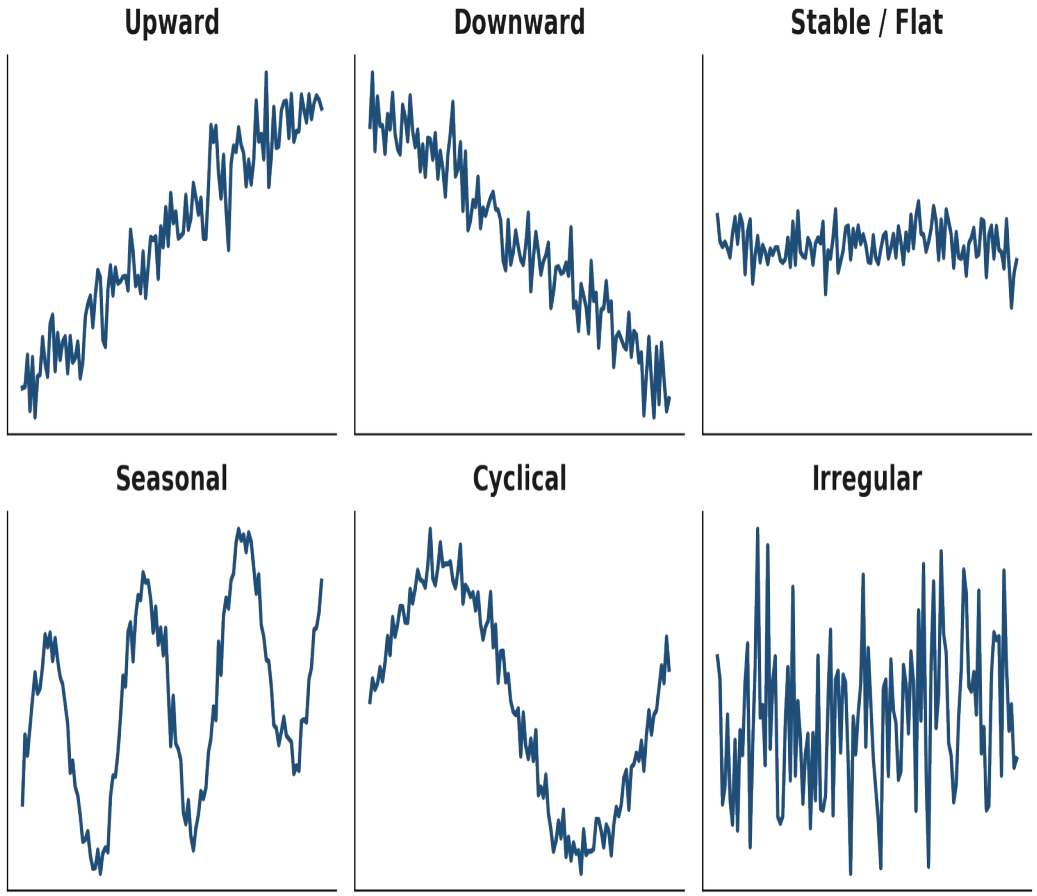
days = [1, 2, 3, 4, 5, 6, 7]
temps = [30, 32, 31, 33, 35, 34, 36]

plt.plot(days, temps)
plt.xlabel("Day")
plt.ylabel("Temperature (C)")
plt.title("Temperature Over a Week")
plt.show()
```

CONCEPT

Types of Trends

- Upward and downward trends show values generally rising or falling over time
- A stable (flat) trend shows no clear rise or fall, just noise around a steady level
- Seasonal trends repeat at a regular, predictable interval, like sales spiking every summer
- Cyclical trends also repeat, but without a fixed period; irregular data shows no pattern at all



CONCEPT

Univariate Analysis: One Variable at a Time

- Univariate means examining a single column on its own, with nothing else in the picture
- For a numeric column: a histogram shows the shape of the distribution, a box plot shows spread and outliers
- For a categorical column: a bar chart of counts shows which categories are most common
- Typical univariate questions: what's the average, how spread out are the values, are there outliers

CONCEPT

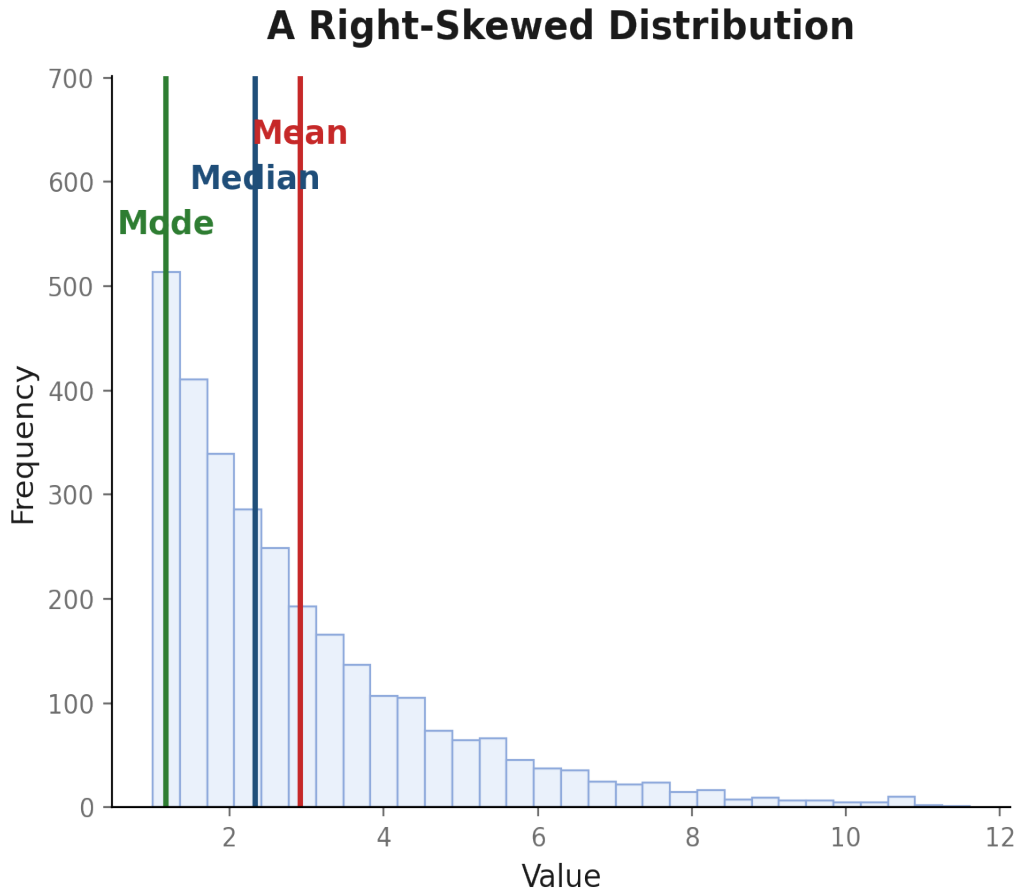
Why Check Central Tendency?

- A single number like the average lets you summarize thousands of rows into one comparable figure
- Comparing the central tendency across groups is often the first real finding in an analysis, like which city sells more
- Choosing the wrong measure, mean on skewed data, can quietly mislead a decision without looking wrong
- It's usually the very first statistic anyone asks for about a new dataset

CONCEPT

Central Tendency: Mean, Median & Mode

- Mean is the average: add every value and divide by how many there are
- Median is the middle value when all values are sorted; it ignores extreme outliers that pull the mean around
- Mode is the most frequently occurring value, the only one of the three that works on categorical data too
- On this skewed chart, the long tail of high values pulls the mean to the right of the median, and past the mode



EXAMPLE

Mean, Median & Mode in Pandas

Each measure answers a slightly different question about the same column.

```
import pandas as pd

scores = pd.Series([55, 60, 62, 70, 72, 75, 78, 95])

print(scores.mean())    # 70.875, pulled up by the 95
print(scores.median())  # 71.0, the middle value
print(scores.mode())    # no repeats here, so every value appears once
```

EXAMPLE

Mean, Median or Mode: Which One?

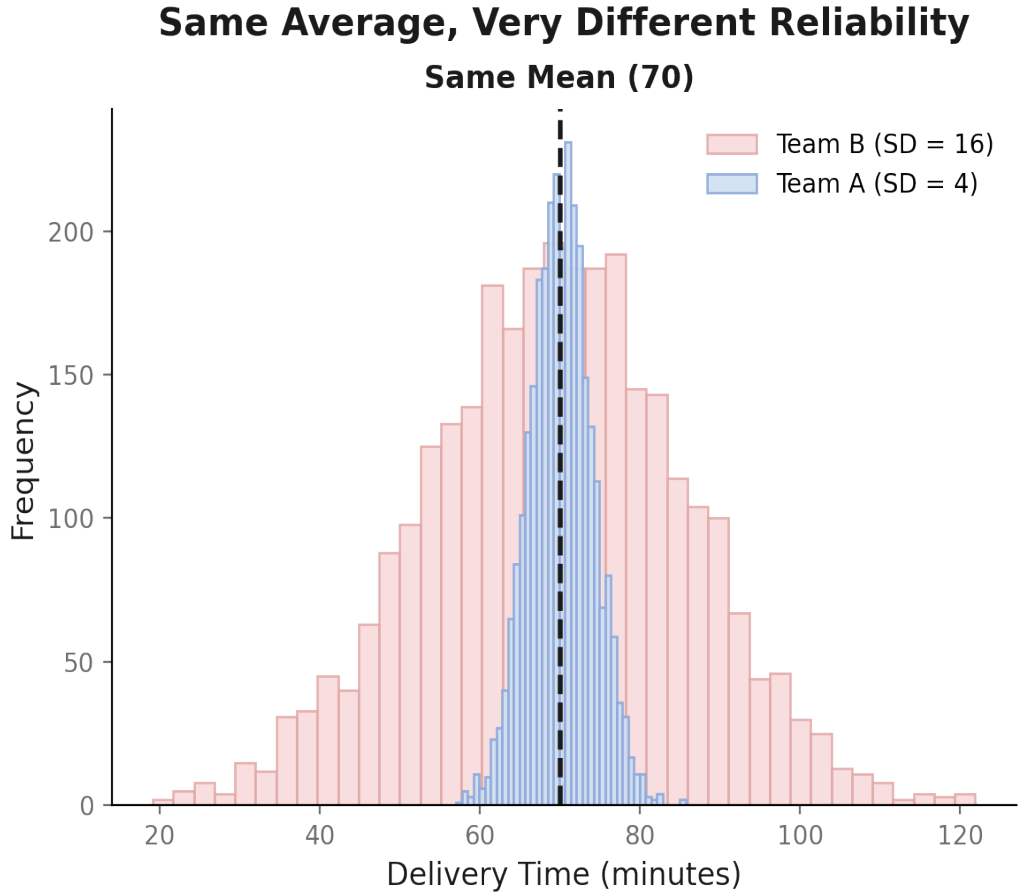
The right measure depends on the data's shape and type, not personal preference.

Measure	Best When	Real Example
Mean	Numeric data, roughly symmetric, no extreme outliers	Average test score in a class
Median	Numeric data that's skewed or has outliers	Median household income, median home price
Mode	Categorical data, or the most common value matters	Most popular shoe size sold, most common survey answer

CONCEPT

Why Check Spread?

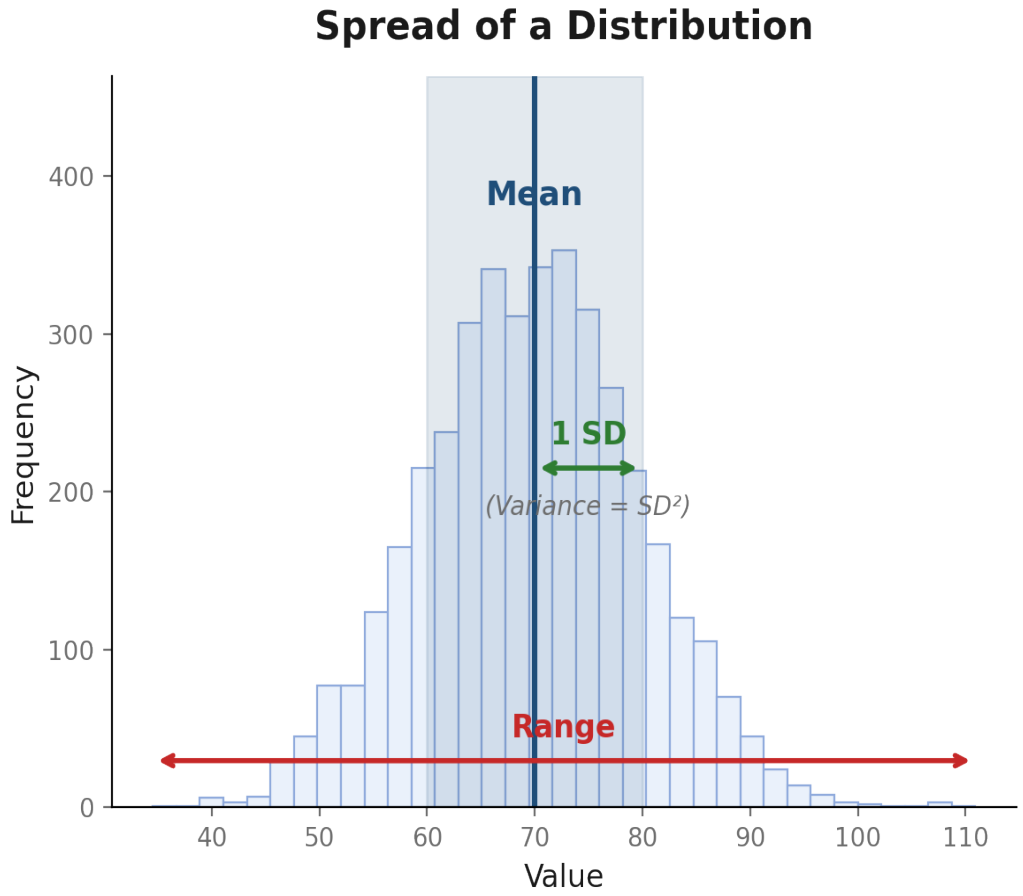
- Two datasets can share the exact same average and still look completely different up close
- Spread tells you how trustworthy that average actually is: tightly clustered data makes the mean reliable, scattered data doesn't
- In real decisions, spread often matters as much as the average: a consistent delivery time beats a good average with huge swings
- Ignoring spread is how outliers and real risk get hidden behind one reassuring number



CONCEPT

Spread: Range, Variance & Standard Deviation

- Range is the simplest measure of spread: the maximum value minus the minimum, shown by the outer arrow
- Standard deviation (SD) marks how far values typically sit from the mean, shown by the shaded 1 SD band
- Variance is standard deviation squared, the same spread in squared units instead of the original ones
- A small standard deviation means values cluster tightly around the mean; a large one means they're spread out



EXAMPLE

Range, Variance & Standard Deviation in Pandas

std() is the one you'll reach for most often, since its units match the original column.

```
import pandas as pd

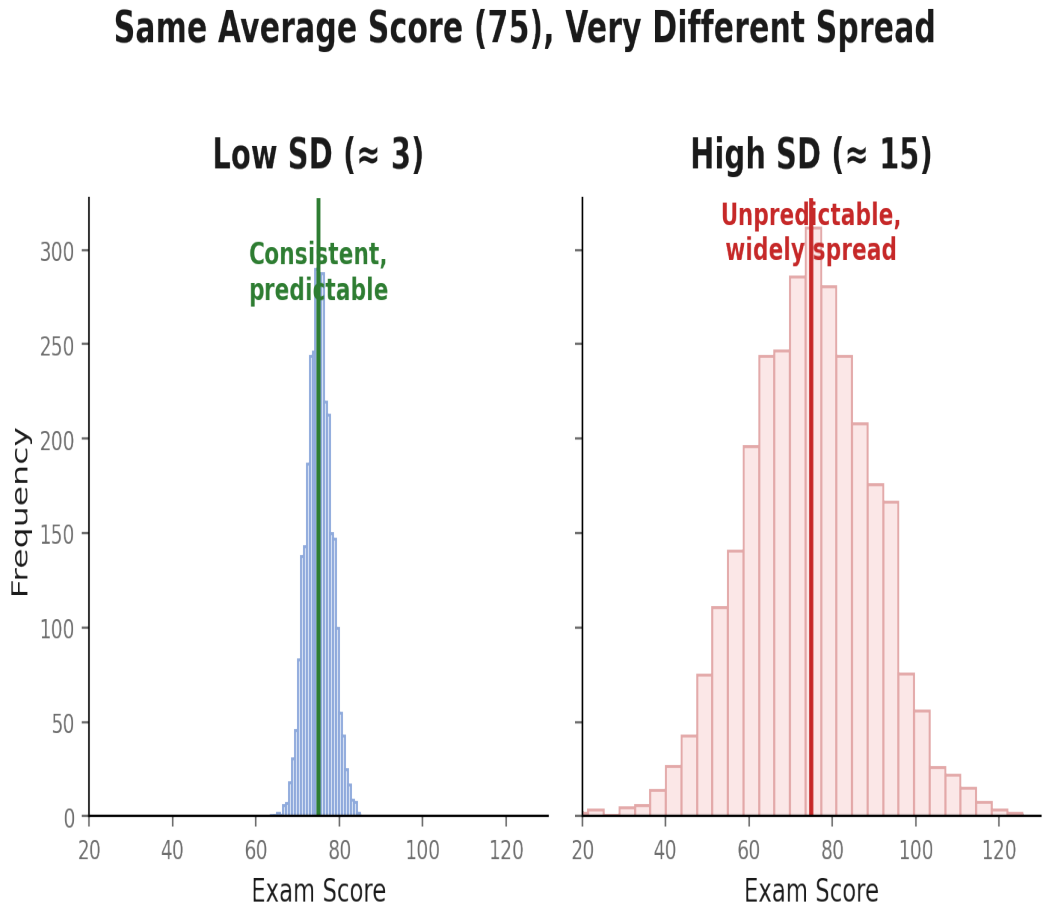
scores = pd.Series([55, 60, 62, 70, 72, 75, 78, 95])

print(scores.max() - scores.min())  # range: 40
print(scores.var())                 # variance, in squared units
print(scores.std())                 # standard deviation: about 12.3
```

CONCEPT

What Low vs. High SD Means in Real Life

- A low SD means values cluster close to the average: predictable, consistent, low risk
- A high SD means values swing far from the average: variable, less predictable, higher risk
- Example: a factory machine with low SD on part size is in control; high SD means inconsistent, defective parts
- Example: a stock with low SD in daily returns is stable; a stock with high SD is volatile and riskier to hold



EXAMPLE

Reading SD in Real Situations

The same average can hide very different realities once you look at the SD.

Situation	Low SD Means	High SD Means
Delivery times	Orders consistently arrive on time	Some orders are fast, others very late
Exam scores	Most students understood the material similarly	Understanding varies widely across students
Stock returns	A stable, lower-risk investment	A volatile, higher-risk investment
Manufacturing	Parts are consistently within spec	Parts vary, more likely to fail quality checks

THINK & PRACTICE

Quick Check

Given ratings = pd.Series([3, 4, 4, 5, 2, 4, 5, 1]), compute and print the mean, median, mode, and standard deviation. In a comment, state whether the ratings are tightly clustered or widely spread.

ANSWER

Quick Check: Answer

```
import pandas as pd

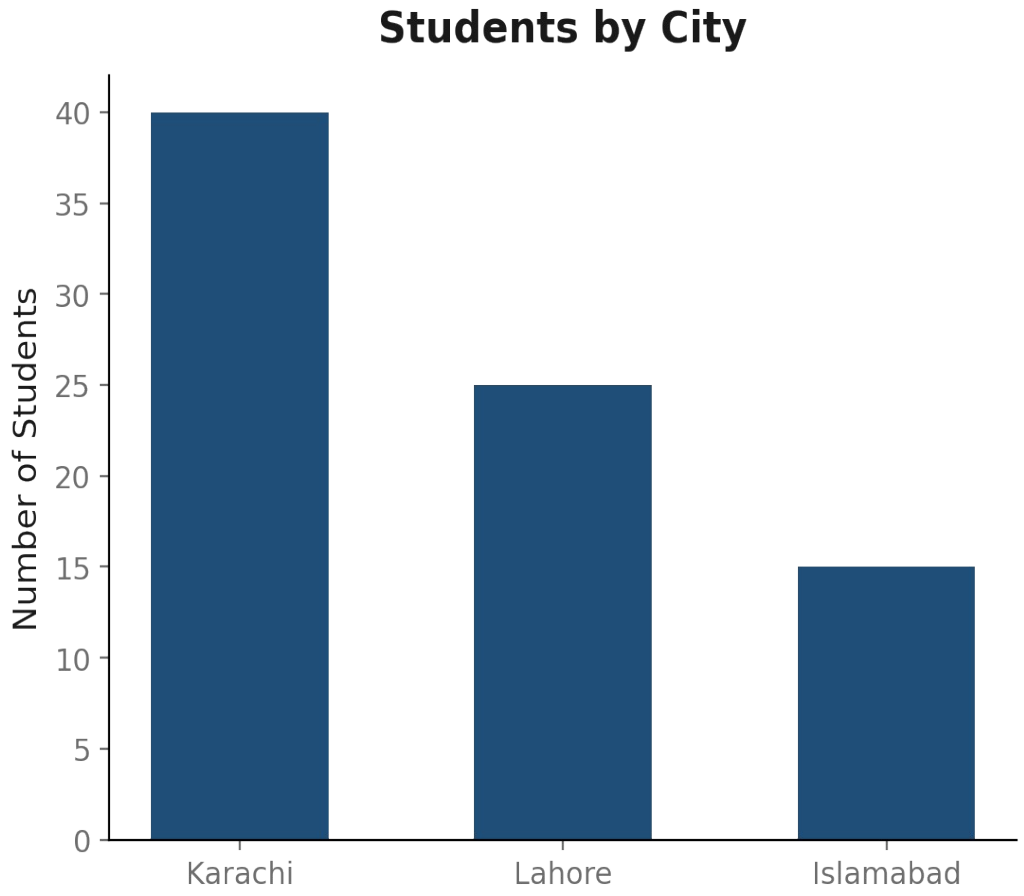
ratings = pd.Series([3, 4, 4, 5, 2, 4, 5, 1])

print(ratings.mean())
print(ratings.median())
print(ratings.mode())
print(ratings.std())
# standard deviation is a bit over 1 on a 1 to 5 scale,
# so ratings are moderately spread, not tightly clustered
```

CONCEPT

Bar Charts: Comparing Categories

- plt.bar(categories, values) draws one bar per category, height equal to its value
- Use a bar chart when you're comparing separate groups, not tracking change over time
- Categories go on the x-axis as labels, not numbers
- A horizontal version, plt.barh(), helps when category labels are long



EXAMPLE

Students by City

Each city becomes one labeled bar.

```
import matplotlib.pyplot as plt

cities = ["Karachi", "Lahore", "Islamabad"]
counts = [40, 25, 15]

plt.bar(cities, counts)
plt.xlabel("City")
plt.ylabel("Number of Students")
plt.title("Students by City")
plt.show()
```

THINK & PRACTICE

Quick Check

Plot a bar chart of average score by subject: subjects = ["Math", "Science", "English"], averages = [78, 85, 90]. Label both axes and add a title.

ANSWER

Quick Check: Answer

```
import matplotlib.pyplot as plt

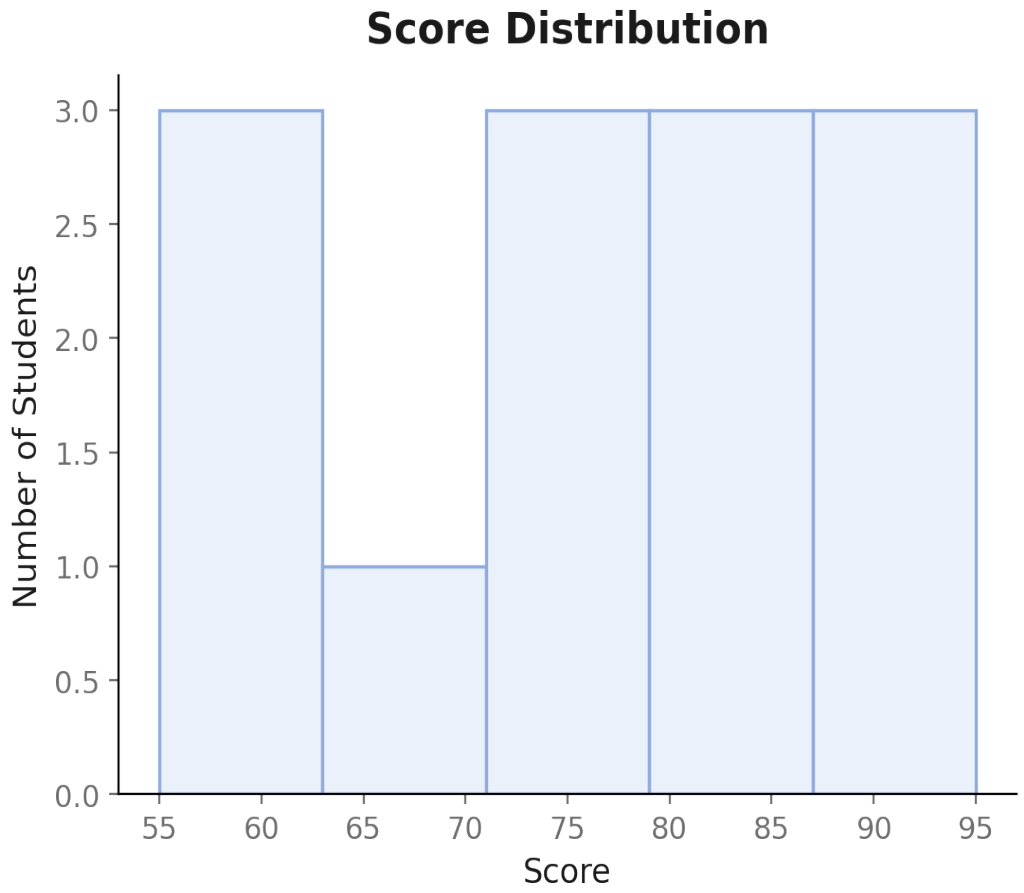
subjects = ["Math", "Science", "English"]
averages = [78, 85, 90]

plt.bar(subjects, averages)
plt.xlabel("Subject")
plt.ylabel("Average Score")
plt.title("Average Score by Subject")
plt.show()
```

CONCEPT

Histograms: The Shape of One Variable

- plt.hist(values, bins=n) groups numbers into n ranges (bins) and counts how many fall in each
- A histogram shows the distribution of a single numeric column: where most values cluster, and where they spread out
- Unlike a bar chart, the x-axis here is a continuous range, not separate categories
- More bins show finer detail; too many bins can make the shape look noisy



EXAMPLE

Distribution of Exam Scores

Five bins group the 13 scores into 5 ranges and count how many fall in each.

```
import matplotlib.pyplot as plt

scores = [55, 60, 62, 70, 72, 75, 78, 80, 82, 85, 88, 90, 95]

plt.hist(scores, bins=5)
plt.xlabel("Score")
plt.ylabel("Number of Students")
plt.title("Score Distribution")
plt.show()
# most scores cluster in the middle-to-upper range
```

THINK & PRACTICE

Quick Check

Given ages = [18, 19, 19, 20, 21, 21, 21, 22, 23, 25, 28, 30], plot a histogram with 4 bins. Label the axes and title it "Age Distribution".

ANSWER

Quick Check: Answer

```
import matplotlib.pyplot as plt

ages = [18, 19, 19, 20, 21, 21, 21, 22, 23, 25, 28, 30]

plt.hist(ages, bins=4)
plt.xlabel("Age")
plt.ylabel("Count")
plt.title("Age Distribution")
plt.show()
```

15-Minute Break

Stretch, grab a coffee. We'll resume shortly.

Hour 2

Bivariate Analysis

CONCEPT

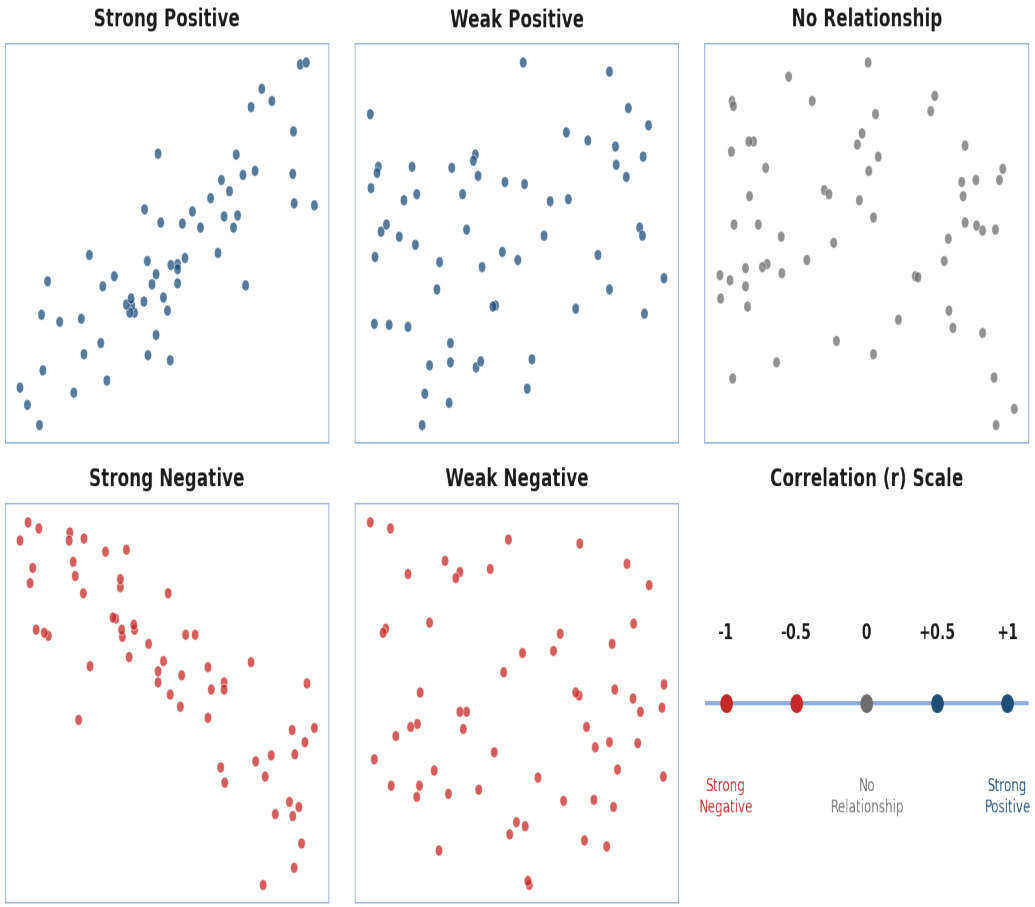
Bivariate Analysis: Looking for a Relationship

- Bivariate means comparing exactly two variables to see whether they move together
- Two numeric columns: a scatter plot or regplot shows the shape, .corr() gives a number for the strength
- A numeric column across a categorical one: a box plot per category compares them side by side
- Typical bivariate questions: does X predict Y, which category has the higher average

CONCEPT

Types of Relationships

- Positive: as one variable goes up, the other tends to go up too (hours studied and score)
- Negative: as one variable goes up, the other tends to go down (price and units sold)
- No relationship: the points scatter randomly, one variable gives no hint about the other
- Strong vs. weak describes how tightly the points hug a line; r close to -1 or +1 is strong, r close to 0 is weak



EXAMPLE

Bivariate Analysis: Which Plot to Use?

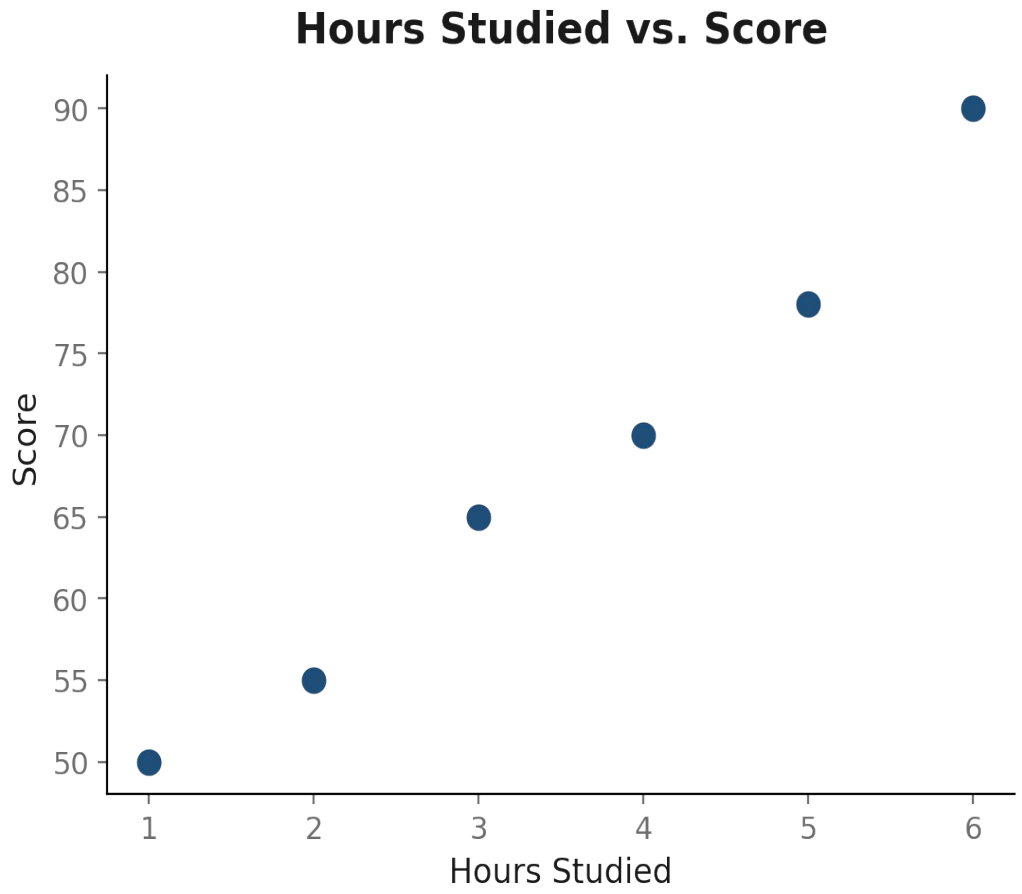
The right chart depends on whether each variable is numeric or categorical.

Variable Pair	Best Plot	Example Question
Numeric vs. Numeric	Scatter plot or regplot	Does hours studied predict score?
Numeric vs. Categorical	Box plot per category	Does average score differ by city?
Categorical vs. Categorical	Grouped bar chart or crosstab	Does city relate to pass or fail rate?

CONCEPT

Scatter Plots: Two Variables at Once

- plt.scatter(x, y) draws one point per pair of values, no lines connecting them
- Use a scatter plot to look for a relationship between two numeric variables
- If the points trend upward left to right, that's a positive relationship; downward is negative
- A cloud with no visible pattern suggests little or no relationship between the two variables



EXAMPLE

Hours Studied vs. Score

The points trend upward: more hours studied tends to come with a higher score.

```
import matplotlib.pyplot as plt

hours_studied = [1, 2, 3, 4, 5, 6]
score = [50, 55, 65, 70, 78, 90]

plt.scatter(hours_studied, score)
plt.xlabel("Hours Studied")
plt.ylabel("Score")
plt.title("Hours Studied vs. Score")
plt.show()
```

THINK & PRACTICE

Think & Practice (Harder)

Given daily_temp = [20, 22, 25, 28, 30, 33, 35] and ice_cream_sales = [100, 120, 150, 200, 240, 300, 340], make a scatter plot with temperature on the x-axis and sales on the y-axis. In a comment, describe the relationship you'd expect to see.

ANSWER

Think & Practice (Harder): Answer

```
import matplotlib.pyplot as plt

daily_temp = [20, 22, 25, 28, 30, 33, 35]
ice_cream_sales = [100, 120, 150, 200, 240, 300, 340]

plt.scatter(daily_temp, ice_cream_sales)
plt.xlabel("Temperature (C)")
plt.ylabel("Ice Cream Sales")
plt.title("Temperature vs. Ice Cream Sales")
plt.show()
# a clear positive relationship: hotter days, more sales
```

CONCEPT

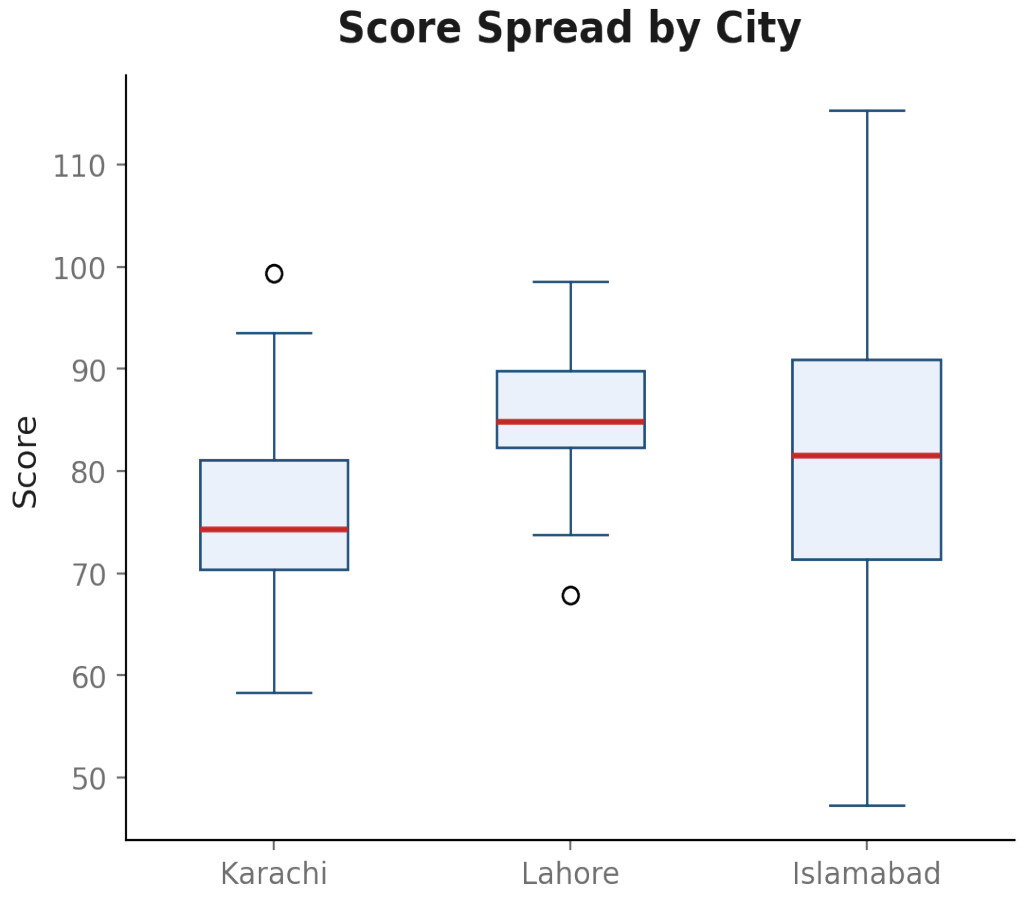
Why Seaborn?

- Seaborn is built directly on top of Matplotlib, with nicer defaults and statistical plots included
- Seaborn functions usually take a DataFrame directly: sns.boxplot(x='city', y='score', data=df)
- It adds plot types Matplotlib doesn't have built in, like box plots and correlation heatmaps
- You still finish with plt.show(), since Seaborn draws through Matplotlib underneath

CONCEPT

Box Plots: Spread, Median & Outliers

- A box plot shows the median (middle line), the middle 50% of values (the box), and the overall spread (the whiskers)
- Points drawn outside the whiskers are flagged as possible outliers
- Box plots are ideal for comparing a numeric column across several categories side by side
- Where a histogram shows one distribution's shape, a box plot compares several at once



EXAMPLE

Score Spread by City

One box per city, so you can compare medians and spread at a glance.

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df = pd.DataFrame({
    "city": ["Karachi", "Karachi", "Lahore", "Lahore", "Islamabad"],
    "score": [70, 85, 90, 95, 88],
})

sns.boxplot(x="city", y="score", data=df)
plt.title("Score Spread by City")
plt.show()
```

THINK & PRACTICE

Quick Check

Given a DataFrame df with columns {subject, score} covering Math and Science scores for several students, draw a box plot with subject on the x-axis and score on the y-axis.

ANSWER

Quick Check: Answer

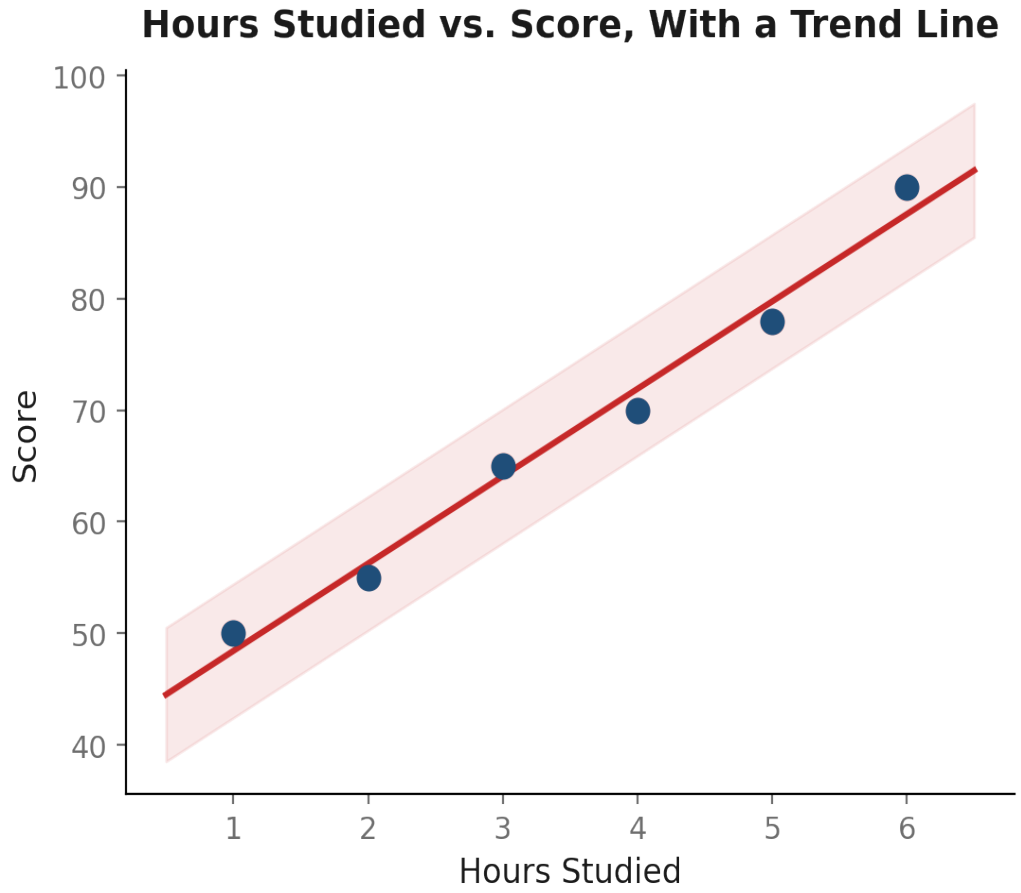
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(x="subject", y="score", data=df)
plt.title("Score by Subject")
plt.show()
```

CONCEPT

Reading a Relationship from a Chart

- A scatter plot shows the shape of a relationship; the correlation number confirms how strong it is
- sns.regplot() draws a scatter plot with a best-fit line through it, making the trend easier to see
- A correlation near +1 or -1 is a strong relationship; near 0 means the two variables barely move together
- Correlation is not proof of cause: two variables can move together without one causing the other



WORKED EXAMPLE

Hours Studied vs. Score, With a Trend Line

regplot adds a fitted line on top of the scatter; the correlation number confirms the strength.

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df = pd.DataFrame({
    "hours_studied": [1, 2, 3, 4, 5, 6],
    "score": [50, 55, 65, 70, 78, 90],
})

sns.regplot(x="hours_studied", y="score", data=df)
plt.title("Hours Studied vs. Score")
plt.show()

print(df["hours_studied"].corr(df["score"]))
# close to 1: a strong positive relationship
```

THINK & PRACTICE

Practice

Given attendance = [60, 65, 70, 80, 90, 95] and score = [55, 58, 68, 75, 88, 92] in a DataFrame, draw a regplot of attendance vs. score, print their correlation with .corr(), and state in a comment whether the relationship looks strong or weak.

ANSWER

Practice: Answer

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df = pd.DataFrame({
    "attendance": [60, 65, 70, 80, 90, 95],
    "score": [55, 58, 68, 75, 88, 92],
})

sns.regplot(x="attendance", y="score", data=df)
plt.title("Attendance vs. Score")
plt.show()

print(df["attendance"].corr(df["score"]))
# strong positive relationship, correlation close to 1
```

THINK & PRACTICE

Practice (Harder)

Given shoe_size = [7, 8, 6, 9, 7, 8] and typing_speed = [45, 60, 50, 40, 55, 48] in a DataFrame, draw a regplot of shoe_size vs. typing_speed and print their correlation. In a comment, explain why a low correlation here doesn't mean your code is wrong, it means these two variables genuinely aren't related.

ANSWER

Practice (Harder): Answer

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df = pd.DataFrame({
    "shoe_size": [7, 8, 6, 9, 7, 8],
    "typing_speed": [45, 60, 50, 40, 55, 48],
})

sns.regplot(x="shoe_size", y="typing_speed", data=df)
plt.show()

print(df["shoe_size"].corr(df["typing_speed"]))
# close to 0: not every pair of columns has a real
# relationship, and that's a correct, useful finding
```

Hour 3

Multivariate Analysis & Styling

CONCEPT

Multivariate Analysis: Several Variables Together

- Multivariate means looking at three or more variables in the same chart or table
- A full correlation heatmap scans every pair of numeric columns in a dataset at once
- Splitting a box plot or scatter plot by color or category adds a third variable to a bivariate chart
- Typical multivariate questions: which combination of factors matters most, does a relationship change across groups

EXAMPLE

Multivariate Analysis: Which Plot to Use?

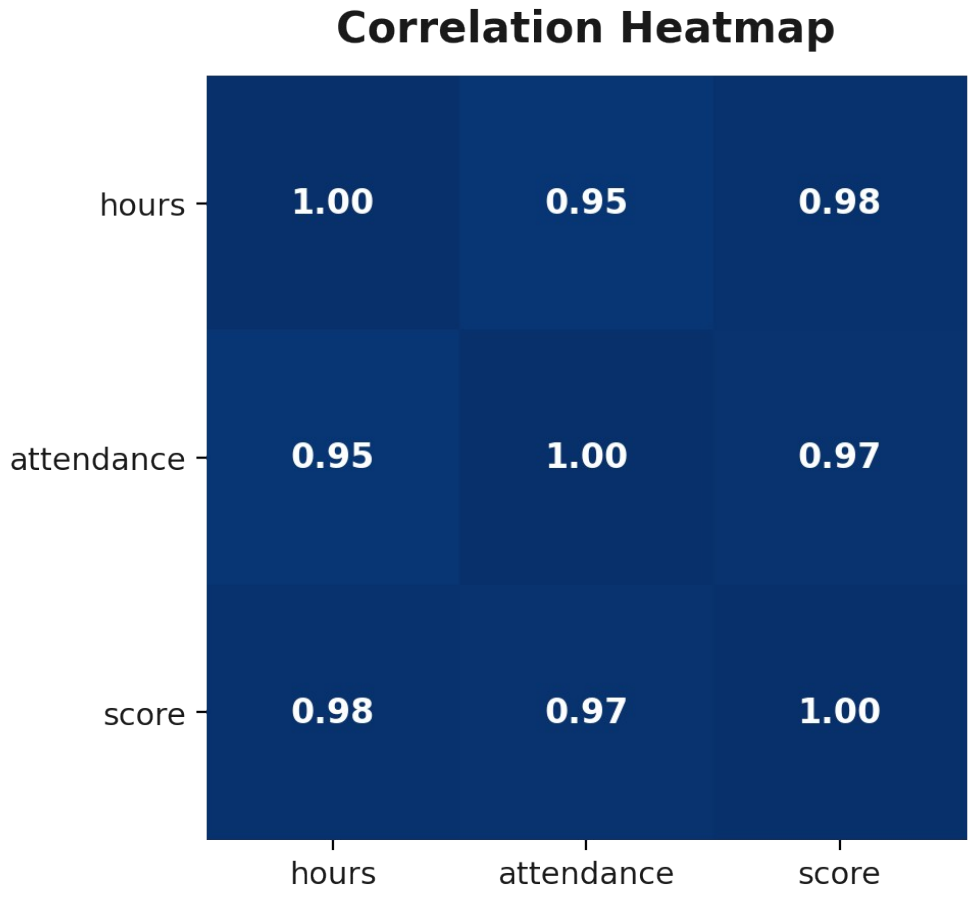
Once a third variable joins the picture, these are the go-to charts.

Situation	Best Plot	Example Question
Many numeric columns at once	Correlation heatmap	Which factors move together across the dataset?
A relationship split by category	Scatter or box plot, colored/grouped by category	Does the hours-vs-score link change by city?
Every pair of numeric columns	sns.pairplot()	Which variable pairs are worth a closer look?
Two categories plus a numeric measure	Grouped bar chart	Does average score differ by city and gender together?

CONCEPT

Correlation Heatmaps

- df.corr() computes the correlation between every pair of numeric columns, from -1 to 1
- 1 means a perfect positive relationship, -1 a perfect negative one, 0 means no linear relationship
- sns.heatmap(corr_matrix, annot=True) draws that grid as colored cells with the numbers printed on top
- This is often the fastest way to scan a whole dataset for interesting relationships at once



EXAMPLE

Correlation Between Study Habits and Score

annot=True prints each correlation value directly on its cell.

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df = pd.DataFrame({
    "hours": [1, 2, 3, 4, 5],
    "attendance": [60, 65, 75, 80, 90],
    "score": [50, 55, 65, 70, 78],
})

corr = df.corr()
sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap")
plt.show()
```

THINK & PRACTICE

Quick Check

Given a DataFrame df with numeric columns {temp, humidity, sales}, compute its correlation matrix with .corr() and draw it as a heatmap with the values annotated on each cell.

ANSWER

Quick Check: Answer

```
import seaborn as sns
import matplotlib.pyplot as plt

corr = df.corr()
sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap")
plt.show()
```

CONCEPT

Styling Seaborn Plots

- sns.set_style('whitegrid') or 'darkgrid' changes the overall look of every plot after that line
- sns.set_palette('pastel') changes the color scheme used for categories
- Styling calls affect every chart drawn afterward, so set them once near the top of your script
- Good styling isn't decoration, it's what makes a chart readable to someone who wasn't in the room when you made it

EXAMPLE

Applying a Style and Palette

The style and palette are set once, then every chart after them picks it up automatically.

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_style("darkgrid")
sns.set_palette("pastel")

sns.barplot(x="city", y="score", data=df)
plt.title("Average Score by City")
plt.show()
```

THINK & PRACTICE

Think & Practice (Harder)

Set the Seaborn style to "whitegrid", then draw a box plot of score by city from the earlier city/score DataFrame, with a clear title. Explain in a comment why you set the style before drawing the plot, not after.

ANSWER

Think & Practice (Harder): Answer

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_style("whitegrid")

sns.boxplot(x="city", y="score", data=df)
plt.title("Score Spread by City")
plt.show()

# styling must be set before drawing; it changes how
# the next plots are rendered, not plots already drawn
```

Home Practice: Interview-Style Questions

- When would you choose a bar chart over a histogram, given the same numbers?
- What does a scatter plot show that a correlation number alone does not?
- Explain what annot=True does in sns.heatmap(), and why it's useful
- Why does 'correlation is not causation' matter when reading a heatmap?
- What's the difference between plt.plot() and plt.scatter(), and when would each mislead you if swapped?
- Given a new dataset, what are the first two or three charts you would draw, and why those?

Key Takeaways & Coming Up

- Univariate charts (bar, histogram) summarize one column; bivariate charts (scatter, box plot, regplot) compare two
- Multivariate tools, like a correlation heatmap, scan several variables for relationships at once
- Seaborn builds on Matplotlib, works directly with DataFrames, and adds box plots and heatmaps
- A chart suggests a relationship; it never proves one variable causes another
- Coming up next: the Month 1 milestone, a full EDA project on a real dataset from start to finish